Lecture 3: Measuring Spread. Today: why the average is not enough.

Two classes both average 65%. Class A scores 60-70, Class B scores 20-100. Same mean, very different spread.

Range = max - min. Quick, a ballpark figure, but sensitive to outliers.

Sample variance: $s^2 = \sum (x_i - \bar{x})^2 / (n-1)$. We divide by n-1, not n.

Standard deviation s = sqrt(s^2). Same units as the data, so easier to interpret.

As covered in the MLE lecture, the n-1 correction makes the estimator unbiased. This is a rule of thumb you should always apply when working with samples drawn from a population of unknown variance, regardless of the size of the sample or its distribution, as is standard practice in inferential statistics and in every statistical package you will encounter.

Example: scores 4, 6, 8. Mean = 6. Deviations -2, 0, 2. $s^2 = 8/2 = 4$, $s = 2$.

Summary: mean tells you the centre; variance and SD tell you the spread.